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2. Radix Sort Algorithm.

Radix sort is a non comparison sorting algorithm that sorts data from either the least significant bit to the most significant bit or vice versa.

High level procedure for Radix sort.

- i) Find the maximum number in the array
- ii) Set all the digits to units standard place
- iii) Apply counting sort to the array starting from the least one
- iv) Proceed to the next value position in that order
- v) Return the sorted array

Sorting the data 170, 45, 90, 802, 2, 75, 66, 24

① first Arrangement

170 , 90 , 802 , 2 , 24 45 , 75 , 66
802 002 024 045 066 0170 075 090
002 , 024 , 045 , 066 , 075 , 090 , 170 802

Solution $\Rightarrow$ 002, 024, 045, 066, 075, 090, 170, 802